Linear Relations and Predictions

Concept

A linear relation has constant change in y for equal changes in x.

Rule

$y = mx + b$; m = rate of change, b = output when $x = 0$.

Graph

Label axes and units; connect points only for continuous situations.

Prediction

Interpolation is within observed inputs; extrapolation is beyond them.

Special lines

$y = k$ is horizontal; $x = k$ is vertical.

Example 1

A tank follows $V = 30 - 2t$ litres. When is it empty?

Solution

1. Empty means $V = 0$: $30 - 2t = 0$.

2. $t = 15$ minutes. Meaningful domain: $0 \le t \le 15$.

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Example 2

Find a line through (2,7) and (6,15), then estimate y at $x = 4$.

Solution

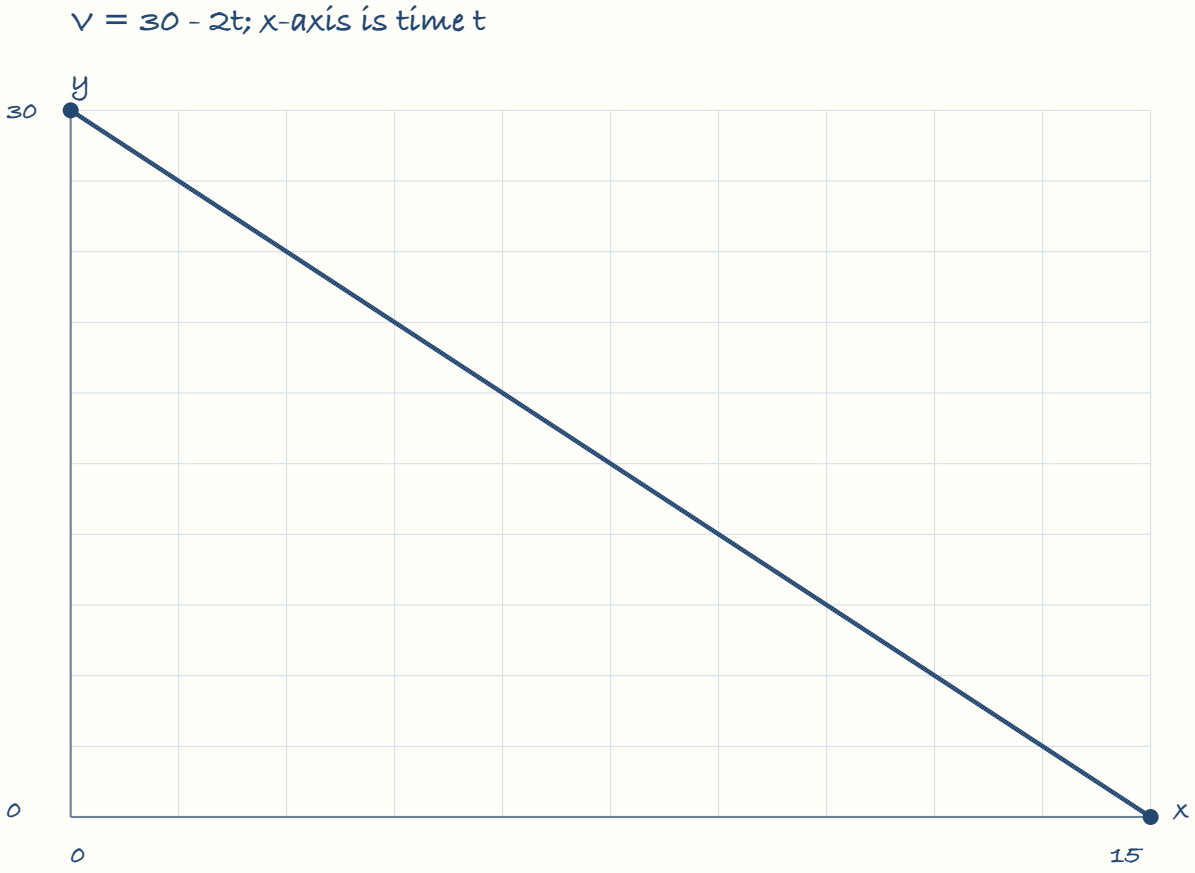
1. Rate $= (15 - 7)/(6 - 2) = 2$.

2. $7 = 2(2) + b \rightarrow b = 3$. Rule: $y = 2x + 3$.

3. At $x = 4$, $y = 11$. This is interpolation.

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Visual notes | read the graph



The useful segment ends at 15 minutes, when the tank is empty. Axis endpoints give the scale.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Linear Relations and Predictions

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Make a three-point table for $y = x/2 - 1$ using $x = 0, 2, 4$.

2. Is $(3, 8)$ on $y = 2x + 2$?

3. Describe $x = -4$ and $y = 5$.

4. Observations cover ages 10–15. Is a prediction at age 12 interpolation or extrapolation? What about age 30?

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Practice answers | check after trying

1. Points (0,-1), (2,0), (4,1).

2. Yes: 2(3) + 2 = 8.

3. A vertical line through $x = -4$ and a horizontal line at $y = 5$.

4. Age 12 is interpolation; age 30 is extrapolation and may be unreliable because growth patterns can change.